

Topología inicial

Definición 1 (Topología inicial). Sean $X \neq \emptyset$, $\{(Y_\alpha, \mathcal{T}_\alpha)\}_{\alpha \in I}$ un conjunto de espacios topológicos y $\mathcal{F} = \{f_\alpha: X \rightarrow Y_\alpha\}_{\alpha \in I}$, $\sigma(X, \mathcal{F})$ es la topología inicial en X inducida por $\mathcal{F} \iff$ es la topología menos fina que hace continuas todas las $f_\alpha \in \mathcal{F}$.

Como la intersección de topologías es una topología, tenemos que

$$\sigma(X, \mathcal{F}) := \bigcap \{ \mathcal{T} \text{ topología en } X : \forall \alpha \in I : f_\alpha: (X, \mathcal{T}) \rightarrow (Y_\alpha, \mathcal{T}_\alpha) \text{ es continua} \}.$$

Referenciado en

- `prop-base-topologia-inicial`
- `teo-abiertos-topologia-inicial`
- `prop-subbase-topologia-inicial`
- `topologia-debil`

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